```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is cPIKAN
Layers: ModuleList(
  (0): ChebyKANLayer(in_features=1, out_features=10, degree=3)
  (1): ChebyKANLayer(in_features=10, out_features=10, degree=3)
  (2): ChebyKANLayer(in_features=10, out_features=1, degree=3)
)
Total trainable parameters: 480
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:09<00:00, 219.67epoch/s, Loss=5.27e-06, LR=1.0e-03, Best=5.27e-06]
Training finished. Best loss: 4.13e-06. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 9.923851013183594s
Relative loss: 0.002294833306223154
L2 loss: 0.001460205647163093
Parameter containing:
tensor([[[ 0.0745,  0.1391,  0.4805, -0.0707],
         [-0.0276, -0.5601, -0.2215,  0.1875],
         [-0.0734,  0.3004,  0.3437,  0.0316],
         [-0.0863, -0.2035, -0.0690,  0.7724],
         [ 0.1894, -0.1528, -0.5296, -0.0884],
         [-0.6191,  0.1519,  0.3496, -0.0694],
         [ 0.2834, -0.0100, -0.2495, -0.2299],
         [ 0.0074,  0.3340, -0.2866, -0.3125],
         [ 0.6931, -0.1862, -0.6569, -0.4033],
         [ 0.3029, -0.3406, -0.5581, -0.0576]]], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[[-6.9197e-03,  2.0064e-01,  1.4386e-02, -1.7630e-01],
         [ 5.1148e-02, -2.1878e-01, -6.8236e-02,  2.3518e-01],
         [ 2.2582e-02,  1.0787e-01, -1.8920e-02, -8.0708e-02],
         [ 1.3016e-02, -2.1806e-01, -2.8725e-02,  2.0010e-01],
         [-2.8419e-02,  1.9678e-01,  4.9081e-02, -1.3530e-01],
         [ 3.8864e-02, -1.7950e-01, -3.4015e-02,  1.7998e-01],
         [ 1.6475e-02, -1.1937e-01, -1.7603e-02,  1.1924e-01],
         [ 5.1629e-02, -2.2025e-01, -1.3762e-02,  1.5817e-01],
         [-2.3872e-02,  1.6204e-01, -1.5072e-02, -2.1201e-01],
         [-2.9759e-02,  2.2888e-01, -7.4038e-03, -1.6150e-01]],

        [[-4.3729e-02, -2.1129e-01,  5.1468e-02,  1.8132e-01],
         [ 3.5696e-02,  2.1906e-01, -8.3999e-03, -1.6950e-01],
         [ 3.7555e-02, -2.2502e-01, -1.0526e-02,  2.4258e-01],
         [ 4.8554e-02,  2.1427e-01, -4.9463e-02, -2.3823e-01],
         [-2.2638e-02, -2.1487e-01,  4.1681e-02,  1.8069e-01],
         [-1.6393e-02,  1.6750e-01, -6.1368e-02, -1.7226e-01],
         [ 4.2585e-02,  1.9374e-01,  2.9361e-03, -1.2565e-01],
         [-1.7974e-02,  2.2411e-01, -2.3223e-02, -2.1395e-01],
         [-6.0101e-03, -2.6491e-01, -3.9928e-03,  2.1097e-01],
         [-6.1339e-02, -2.1172e-01,  6.0602e-02,  1.8714e-01]],
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        [[-2.1242e-02,  1.4827e-01, -2.9056e-02, -1.6713e-01],
         [ 8.6733e-02, -2.8703e-01, -6.2833e-02,  2.7538e-01],
         [ 3.0381e-03,  7.9789e-02,  5.1443e-03, -2.5373e-02],
         [ 2.3368e-02, -2.2786e-01,  4.9103e-03,  2.0383e-01],
         [-8.4490e-02,  2.3809e-01,  6.1119e-02, -1.9144e-01],
         [ 5.1683e-02, -1.3717e-01, -3.1586e-02,  1.9545e-01],
         [ 2.1869e-02, -1.5628e-01, -1.4495e-02,  1.1643e-01],
         [ 3.5718e-02, -2.0807e-01, -3.7630e-02,  1.4940e-01],
         [-4.7026e-02,  1.9544e-01, -3.7285e-02, -1.9114e-01],
         [-7.5684e-03,  1.8094e-01,  1.2289e-02, -1.9489e-01]],

        [[-4.7308e-02, -6.8682e-02, -2.9486e-02, -9.0022e-03],
         [ 9.6810e-02, -5.9428e-02,  1.8027e-02, -8.2072e-02],
         [ 8.6022e-02, -1.5531e-01,  2.7867e-02,  1.2388e-01],
         [-1.9708e-03,  1.0532e-01,  9.9539e-03, -8.1835e-02],
         [-4.4934e-02, -2.6658e-04,  4.9589e-03,  2.3504e-02],
         [ 3.6095e-02, -1.3900e-02, -2.5104e-02, -2.1631e-02],
         [-6.1381e-03, -9.9824e-03, -3.2302e-02, -9.4130e-03],
         [ 2.6350e-02,  1.3966e-01, -6.0038e-02, -1.7782e-01],
         [-2.0075e-02, -7.2557e-03,  4.3657e-02,  1.3503e-02],
         [-1.3275e-02, -4.1241e-02,  6.0912e-02,  4.5032e-02]],

        [[-1.9338e-02, -6.7209e-02, -4.7188e-03,  3.9358e-02],
         [-3.7715e-03,  1.3891e-01, -2.3927e-02, -1.3160e-01],
         [ 4.9127e-03, -8.1692e-03, -3.3358e-02, -4.0596e-02],
         [ 8.6339e-03,  1.6699e-02, -8.9813e-03, -4.0989e-02],
         [-3.8764e-02, -5.9773e-02,  3.2322e-03,  7.9966e-02],
         [ 7.6636e-03,  1.1829e-01, -2.2023e-02, -2.9747e-02],
         [ 7.8571e-03,  2.8003e-02, -1.7040e-02, -8.4649e-02],
         [ 8.0942e-03,  8.3657e-02, -4.1891e-02, -6.2213e-02],
         [ 1.9108e-03, -7.6113e-02, -1.0195e-02, -7.0025e-03],
         [-2.8307e-02, -6.7732e-02, -1.6286e-02,  8.2365e-02]],

        [[-3.1779e-02,  4.3909e-02, -2.5524e-02, -2.0603e-02],
         [ 4.8089e-02, -2.9261e-02,  6.4875e-02,  6.7882e-02],
         [ 5.4753e-03, -1.4329e-02, -6.9369e-02,  9.5722e-03],
         [ 1.4536e-02, -8.5932e-02,  7.8903e-02, -2.2186e-02],
         [-2.4184e-02,  1.5973e-02, -3.0140e-02, -6.4681e-02],
         [ 5.0111e-02, -2.2644e-03,  6.0012e-02,  3.6425e-02],
         [ 2.4345e-02, -3.0100e-02,  1.6078e-02, -1.3853e-02],
         [ 2.0498e-02, -5.7007e-02,  1.1445e-01, -2.3995e-02],
         [-6.2842e-03,  1.7920e-02, -1.1737e-02, -1.8524e-02],
         [-4.8484e-02,  9.6246e-03, -4.7952e-02, -1.7922e-02]],

        [[ 3.0047e-02, -1.1536e-02,  6.9839e-02,  3.7801e-02],
         [ 3.5560e-02,  5.5325e-02,  2.0264e-02, -1.0212e-01],
         [-5.9578e-02,  3.7713e-02,  2.4250e-02,  3.9064e-03],
         [-1.7599e-02,  1.4122e-02, -2.9396e-02, -1.6781e-02],
         [-1.0336e-02, -2.9795e-02,  4.8697e-03, -1.1077e-02],
         [ 7.3130e-03,  2.8427e-02,  1.2655e-03, -2.5208e-02],
         [ 5.3464e-02,  3.2356e-02,  1.1475e-02, -4.4908e-02],
         [ 4.5462e-02, -2.5609e-03, -7.8001e-02, -4.2563e-02],
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         [ 5.4453e-03, -7.8145e-03,  4.9463e-02, -1.3551e-03],
         [-6.5846e-02, -3.5482e-02,  2.0587e-02,  1.1297e-02]],

        [[-9.8072e-02, -2.5826e-02,  2.2129e-02,  4.8577e-02],
         [ 5.3082e-02,  1.0177e-01, -1.2450e-02, -2.7389e-02],
         [-1.0057e-02,  2.5237e-02,  4.1042e-02, -7.6389e-02],
         [ 7.4555e-02, -3.7986e-02, -6.4955e-02,  1.4911e-02],
         [-4.4316e-02,  4.9760e-02,  5.2259e-02,  5.9499e-02],
         [ 2.3262e-02,  2.9605e-02, -1.1859e-02, -1.7188e-02],
         [-1.4482e-02,  5.0593e-02, -1.5326e-02, -1.1330e-02],
         [ 4.3177e-02, -3.0504e-02, -5.3239e-02,  2.7202e-02],
         [-5.1382e-02,  1.6547e-03,  4.0360e-02,  1.6669e-02],
         [-5.0380e-02, -3.9542e-02,  6.9444e-03, -1.4787e-03]],

        [[ 1.8718e-02, -3.0910e-02, -1.3432e-01, -1.2445e-02],
         [ 4.7749e-02,  9.4513e-02,  3.1094e-01, -6.5588e-03],
         [-2.1259e-03,  5.7221e-02,  7.0622e-02,  1.3112e-02],
         [-2.9924e-03,  1.0550e-02,  9.4247e-02, -5.2940e-02],
         [-2.9913e-02, -4.8530e-02, -2.1538e-01,  3.8074e-03],
         [-1.3740e-02,  3.2748e-02,  1.5001e-01, -2.9130e-02],
         [ 4.4070e-02, -3.6267e-03,  1.9584e-01, -6.5228e-03],
         [ 4.2635e-02,  3.3968e-02,  7.3081e-02, -9.0924e-02],
         [ 2.9455e-02, -1.7362e-02, -9.8602e-02,  3.4794e-02],
         [-2.2409e-02, -5.4451e-02, -1.5070e-01,  5.6007e-02]],

        [[-9.9124e-03, -2.7751e-02,  4.3649e-02,  6.9771e-02],
         [ 3.6734e-02,  1.1974e-01,  1.1238e-02, -1.4061e-01],
         [ 3.1489e-02,  1.7686e-02, -4.3281e-02, -2.0147e-02],
         [ 1.7429e-02,  7.3412e-02,  5.1482e-02, -3.9884e-02],
         [-2.6065e-02, -5.3426e-02, -1.5235e-02,  5.2642e-02],
         [ 4.2304e-02,  5.6198e-02,  1.1902e-03, -6.3505e-02],
         [ 4.9907e-02,  3.7820e-02, -1.4717e-02, -2.8857e-02],
         [ 3.5511e-02,  9.0798e-02,  1.5426e-01, -2.6070e-02],
         [-8.7316e-02, -5.3074e-02, -4.6433e-02,  3.4425e-02],
         [-2.0290e-02, -9.0144e-02, -2.0412e-02,  4.4562e-02]]],
       device='cuda:0', requires_grad=True)
Parameter containing:
tensor([[[-0.0040, -0.0183, -0.0291,  0.1334]],

        [[ 0.0049,  0.0267, -0.0275, -0.0710]],

        [[ 0.0343,  0.0656,  0.0199, -0.1060]],

        [[ 0.0664,  0.0065, -0.0820, -0.0772]],

        [[-0.0143, -0.0045, -0.0006,  0.0671]],

        [[ 0.0355,  0.0174, -0.0223, -0.0974]],

        [[ 0.0080,  0.0062,  0.0198, -0.0575]],

        [[ 0.0091, -0.0225, -0.0488, -0.0939]],

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```
        [[ 0.0554,  0.0038, -0.0502,  0.0477]],

        [[-0.0261, -0.0235, -0.0183,  0.0519]]], device='cuda:0',
       requires_grad=True)
==========================================================================================
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================================
CPIKANModel                              [1, 1]                    --
├─ModuleList: 1-1                          --                        --
│    └─0.weight                                                      ├─40
│    └─1.weight                                                      ├─400
│    └─2.weight                                                      └─40
│    └─ChebyKANLayer: 2-1                  [1, 10]                   40
│    │    └─weight                                                    └─40
│    └─ChebyKANLayer: 2-2                  [1, 10]                   400
│    │    └─weight                                                    └─400
│    └─ChebyKANLayer: 2-3                  [1, 1]                    40
│    │    └─weight                                                    └─40
==========================================================================================
Total params: 480
Trainable params: 480
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
==========================================================================================
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
==========================================================================================

进程已结束，退出代码为 0

```